2.5.1 Title, researchers and institution

קול קורא 34/2026 · משרד האנרגיה והתשתיות · יחידת המדען הראשי · מסלול אקדמיה

JouleFlex: Energy-Optimal Power Control for AI Data Centers

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2.5.2 Abstract / תקציר

English (300 words)

In July 2026 Israel's Electricity Authority froze new data-center connection requests after the queue reached 27,000 megawatts. Under electricity limits, data centers switch GPUs (AI processors) off or cap their power, deciding in watts, the momentary draw, not joules, the energy consumed. A capped GPU computes slower and pays its fixed power floor longer, so energy per unit of useful work can rise even as the meter shows fewer watts: less AI per kilowatt-hour. A similar challenge exists in aviation, where it is solved: fuel flow rises far faster than speed, yet slow flight burns fuel longer, so airlines cruise at the speed minimizing fuel per kilometer.

A recent PI study measured this for AI (5,500 measurements, 20 models, four GPUs). Energy per task is U-shaped in the power cap. Its minimum, the Joule Point, sits at 43 to 46 percent of rated power. Holding a card there cut energy per task 29 to 31 percent, runs 1.2 times slower. A fleet simulation on the measured curves (only scheduling simulated) served equal work with 18 to 45 percent less energy; a fixed GPU power budget delivered about 50 percent more work per megawatt.

These insights reshape how AI data centers are planned, managed, and priced, raising decisions no tool supports: how to build a fleet whose cards run near their Joule Points; how to measure and predict the Joule Point for new cards and AI jobs; and how to price service to motivate customers, not today's per-hour billing that rewards the energy-worst point. Each is a modeling problem, and this project will build the decision-support toolkit that answers them: an open dataset, models to characterize and predict the Joule Point for any card and workload, models for fleet design and operation near the Joule Point, and grid and pricing interfaces.

עברית (300 מילים)

ביולי 2026 הקפיאה רשות החשמל בקשות חיבור חדשות של מרכזי נתונים לאחר שהתור הגיע ל-27,000 מגה-ואט. כאשר אספקת החשמל מוגבלת, מרכזי נתונים מכבים מעבדי GPU (מעבדי בינה מלאכותית) או קובעים להם תקרת הספק. החלטות אלה מתקבלות בוואטים, המבטאים הספק רגעי, ולא בג'ולים,

המבטאים את האנרגיה הנצרכת. מעבד GPU תחת תקרת הספק מחשב לאט יותר ופועל זמן רב יותר תוך צריכת הספק בסיס קבוע; לכן האנרגיה ליחידת חישוב מועיל עלולה לעלות אף שהמונה מציג פחות ואטים, כלומר פחות עבודה שימושית לכל קילוואט-שעה. אתגר דומה קיים בתעופה האזרחית, ושם הוא נפתר: קצב צריכת הדלק עולה מהר בהרבה מן המהירות, אך בטיסה איטית נצרך דלק במשך זמן רב יותר. לכן חברות התעופה טסות במהירות הממזערת את צריכת הדלק לקילומטר.

מחקר שנערך לאחרונה בידי החוקר הראשי בחן כמותית את יחסי הגומלין האלה במחשוב בינה מלאכותית (5,500 מדידות, 20 מודלים, ארבעה מעבדי GPU). נמצא כי האנרגיה למשימה כתלות בתקרת ההספק מתארת עקומת U. המינימום, נקודת הג׳ול (Joule Point), נמצא ב-43 עד 46 אחוזים מההספק הנקוב. הפעלת מעבד GPU בנקודה זו הפחיתה את האנרגיה למשימה ב-29 עד 31 אחוזים, כאשר זמן הריצה גדל פי 1.2. סימולציית צי על בסיס העקומות שנמדדו (רק התזמון מדומה) סיפקה אותה כמות עבודה בצריכת אנרגיה נמוכה ב-18 עד 45 אחוזים; תקציב הספק קבוע ל-GPU סיפק כ-50 אחוזים יותר עבודה למגה-ואט.

תובנות אלה משנות את האופן שבו מתוכננים, מנוהלים ומתומחרים מרכזי נתונים לבינה מלאכותית תחת הפעלה מודעת-אנרגיה, ומעלות החלטות שאף כלי אינו תומך בהן: כיצד לבנות צי שכרטיסיו יכולים לפעול סמוך לנקודות הג׳ול; כיצד למדוד ולחזות את נקודת הג׳ול לכרטיסים ועבודות חדשים; וכיצד לתמחר שירות כך שיתמרץ לקוחות, במקום החיוב השעתי המתגמל את ההפעלה הגרועה אנרגטית. כל אחת מהן היא בעיית מידול, והפרויקט יפתח את ערכת התמיכה בהחלטות מבוססת-המודל שתענה עליהן: מסד מדידות פתוח, מודלים לאפיון ולחיזוי נקודת הג׳ול לכל כרטיס ועומס, מודלים לתכנון ולהפעלת צי סמוך לנקודת הג׳ול, וממשקי תמחור ורשת.

2.5.3 Glossary

Term	Meaning
Power	How fast electricity is drawn at a given moment; watts (W) or megawatts (MW). The grid connection is sized by peak power.
Energy	The total electricity consumed: power × time; joules or kilowatt-hours. The bill and the carbon footprint are set by energy.
Power cap	A software limit on a GPU's power draw. Set with one command; takes effect in ~0.2 seconds. The control knob of this project.
TDP	A GPU's full rated power. Running at TDP is today's default.
Joule Point	The power cap at which energy per task is lowest for a given card; measured at 43–46% of TDP on large GPUs.
Operating point	The pair (GPU, power cap) a job runs at.
Power envelope	A limit on a facility's total power draw that can change over time, set by a budget, a price or carbon signal, or a grid instruction.
Power shaping	Keeping a fleet inside a power envelope by capping jobs, like traffic shaping in networks.

ELF	Our released dataset: dense power-cap sweeps of 20 AI models on 4 GPUs, ~5,500 measured rows.
SLO	Service level objective: the deadline or latency a job must meet.
Firm / flexible capacity	Grid connection capacity guaranteed at all times, versus capacity given on condition that the facility reduces power when instructed.
Card multiplier	The extra cards needed to keep total throughput when capping. It equals the latency ratio (~1.2×).

2.5.4 Objectives of the program

2.5.4.1 The objective

The objective is to construct **JouleFlex**: a toolkit of models, software tools and a simulator for the energy management of GPU-rich AI data centers, built on the Joule Point concept, in three layers. **Models:** the power-response law validated on current hardware and whole servers; a job model calibrating the energy-optimal cap per workload; a transfer model for unmeasured GPUs; and a joint job-and-cap assignment method. **Tools:** a characterization kit qualifying a new GPU within the validated hardware domain in hours, and a fleet controller holding a facility inside its power envelope while meeting deadlines. **Simulation:** a digital twin replaying measured curves, evaluating any policy or connection scenario before deployment. All layers are released openly with the calibrating corpus (ELF-2), serving two users: the operator who runs it, and the Ministry that plans, regulates and sizes connections with its numbers. The twin answers the fleet-design question directly: how many cards, of which type, run near the Joule Point behind a given connection.

2.5.4.2 The research question

Can AI data centers operate as controllable electrical loads, each GPU near its energy-optimal cap and the facility inside a changing power limit, so that energy per unit of AI work falls and the power capacity required to serve it shrinks, while admitted work keeps its service promises, under Israeli grid constraints?

Six sub-questions structure the work plan:

1. **Does the law generalize?** The power law and Joule Point were measured on 20 models and 4 GPUs; do they hold on H100/B200-class hardware, production serving software, and shared GPUs?
2. **When is one cap per card enough?** The measured Joule Point was nearly constant per card in the loaded regime; the project maps where that constant breaks and fits a job model there.
3. **Where is the true optimum at the wall?** Board-power measurements miss host overhead, and the grid connection is sized at the wall: what is the Joule Point of the whole node, and of the facility?
4. **Can we skip the measurement?** Can an unmeasured GPU's Joule Point be predicted from published electrical characteristics plus at most one short test, characterizing a new card model in hours?

5. **Can allocation and power be optimized together, in motion?** Choose at once which job runs on which card and at what cap, maximizing served work under a time-varying power limit (price, carbon, grid instruction) with every deadline met; the measured result covers fixed caps and a fixed placement rule; the joint problem is open.
6. **How much flexibility can a facility sell, and who has the incentive?** How much power can be shed on instruction, how fast, for how long, at what service cost, the numbers behind conditional connections; and which pricing rules make offering flexibility profitable, when per-hour billing rewards the energy-worst point?

2.5.5 The proposed research

2.5.5.1 Scientific and technological background

The general problem. AI data centers are the world's fastest-growing electricity consumers, with demand projected to approach 945 TWh by 2030 [3].

The Israeli problem is specific and acute. Israel's grid is small and isolated; in July 2026 the Electricity Authority halted new data-center connection requests for 140 days, after the queue reached about 27,000 MW, roughly three times average national consumption [2]. The binding limit is grid capacity in megawatts: the national need is more AI work from every connected megawatt.

Today's operating decisions think in watts, not joules. Under electrical stress a data center either switches GPUs off, discarding service, or caps GPU power energy-blind, ignoring the total energy each job will now consume; a capped GPU computes slower, so its fixed power floor is paid for more seconds, and capping too deep lowers the watts on the dashboard while raising the joules on the bill. Unconstrained, the default is full rated power, where the last bit of speed carries a steep power surcharge.

Our discovery. The applicant's study [1] mapped, over 20 AI models and four GPU types (~5,500 instrumented measurements, released as the ELF dataset), how energy per task depends on the power cap: board power follows a simple physical law in the computing rate θ,

$$P(\theta) = P_0 + a\,\theta^{\beta}, \quad \text{median } R^2 = 0.99,$$

a fixed floor P_0 plus a superlinear term. Energy per task, $E = P/\theta$, is therefore U-shaped in the cap; its minimum, the **Joule Point**, sits at 43 to 46 percent of rated power on the large GPUs, cutting energy per task by 29 to 31 percent at an exactly priced cost: tasks run about 1.2 times slower, and holding throughput takes that factor more cards. The cap is one software command, settling in about 0.2 seconds, fast enough to follow grid signals; behind a fixed feed the same physics yields about 50 percent more AI tasks per megawatt than running uncapped [1].

2.5.5.2 Review of existing knowledge

Four research lines meet here. *Energy-proportional computing* [4] identified the fixed, work-independent part of server power as the central inefficiency; our floor P_0 measures it for modern GPUs, and the Joule Point is where paying it stops being worthwhile. *Speed-scaling theory* proved that energy is minimized at a sub-maximal speed [5,6,7]; the ELF dataset gives the measured form for GPU inference, where the law holds per operating point and breaks when pooled across batch sizes. *GPU energy systems*, notably Zeus [8] and the ML.ENERGY benchmarks [9,10], search for a good setting per job at runtime; the per-card-constant Joule Point replaces that search with a one-time characterization. *Production context*: cluster studies [11] and public traces [12] supply realistic job arrivals; energy-aware GPU schedulers [13,14] leave each device's power draw at its default; methodology reviews [15] motivate our fixed protocol. Among the cited literature we found no system that keeps an AI fleet inside a time-varying, grid-facing envelope with auditable compliance, or that quantifies AI facilities as providers of firm and flexible capacity.

2.5.5.3 Knowledge gaps

- **The law's reach is unmapped.** The law and Joule Point are established for single jobs at fixed settings on three inference GPUs; their persistence on H100/B200-class hardware, production serving software and shared cards is unknown, and placing the Joule Point on a new GPU still requires a full sweep, since prediction from published characteristics is the study's stated next step [1].
- **The job-dependence structure is unmapped.** The per-card constant holds in the measured loaded regime; outside it the optimum moves, and no model predicts a job's operating point from its features.
- **All published numbers are board numbers.** Energy studies measure board power, but the grid meters the wall; host overhead (CPU, memory, cooling, power-supply losses) shifts the true optimum upward by an unmeasured amount; until it is placed at node level, no megawatt claim is grounded.
- **Allocation and power control are not optimized together in the cited work.** The measured 18–45 percent fleet saving used a fixed budget, a fixed placement rule and caps chosen afterwards; the joint problem, assigning jobs and setting every cap at once under hourly-changing limits, couples discrete scheduling to continuous control through the shared power budget and is unformulated.
- **Flexibility is unquantified.** How much power an AI facility can shed, how fast, for how long, at what cost has never been measured or standardized; Israel's regulator is

rationing connections without the numbers to credit flexible ones.

- **The incentive problem is identified but unsolved.** Per-hour billing makes the renter's cheapest choice the energy-worst point, and a per-joule surcharge is too weak to fix it [1]; which structural instruments work is open.

2.5.5.4 Description of the proposed research

JouleFlex builds one measured foundation and three instruments on it: *first compute the optimal total energy for each job on each card, then control power to achieve it.*

The characterization instrument (WP1, WP2). Extends the ELF corpus to current hardware and production serving software, and adds the two missing measurements: job structure (the per-card constant as null model, a job model where it is not enough) and power above the board, at the RAPL node tier and the BMC PSU-input wall tier. A transfer model and one-test protocol then qualify any new card model within hours, making every (card, job) pair's optimum a computed quantity.

The operating instrument (WP3). An open fleet controller choosing at once which job runs on which card (assignment) and at what power cap (control), over a heterogeneous fleet, from the measured curves, the job model, the job stream with deadlines, and the power envelope. Its rule: give each job the least power that meets its deadline, the Joule Point when the deadline allows, below it only when the envelope forces trade-offs, with the energy cost reported; nothing is energy-blind. Moving a job changes the power available to every other job, so the two decisions cannot be optimized separately; treating them together goes beyond the study's fixed-placement result. A digital twin, replaying the measured curves, evaluates every policy against ground truth before deployment.

The grid instrument (WP4, WP5). Defines and measures the flexibility products a controlled facility can offer (committed envelope, flexibility offer, auditable delivery record), demonstrates them in the Israeli firm-plus-flexible scenario, and evaluates candidate pricing and connection instruments in the twin; everything is released openly, since connection policy cannot rest on a vendor black box.

2.5.5.5 Methods (work packages)

WP	Content
WP1	**ELF-2: the extended measurement corpus, at board and at wall.** The sweep protocol of [1], unchanged, on new GPU generations (H100/L40S class; B200 as available) and production engines (vLLM [16], TensorRT-LLM), covering LLM phases, shared GPUs and fine-tuning, extended into regimes where the per-card constant should break. Three rented measurement tiers: GPU board power (NVML) on AWS EC2 instances with full-GPU passthrough and root access; node power (RAPL plus NVML) on AWS bare metal; PSU input power read out-of-band from the BMC (IPMI

	DCMI / Redfish) on rented dedicated servers, cross-validated once against a calibrated reference meter, placing the Joule Point at the level the grid sees; one fixed protocol throughout.
WP2	**Job model, transfer model, and one-test protocol.** The job model: with the per-card constant as null, predict the per-job optimum from job features (model family, batch, phase, memory pressure), with error bars. The transfer model: predict the law's parameters for unmeasured cards, β from published voltage-frequency behavior, P_0 from one idle measurement, evaluated leave-one-card-model-out on at least five fully swept GPU models spanning three architecture generations, with error reported per held-out model and claims restricted to the validated domain; together they qualify a new card model in hours at known cost. Curvature analysis: the penalty of one cap per card is second order in the dispersion of per-job optima, with constant set by the exponent β, so the measured under-one-percent penalty becomes a proved bound.
WP3	**Joint allocation-and-capping controller, and the digital twin.** Assign jobs to heterogeneous cards and set every cap simultaneously under a time-varying envelope, maximizing served work with hard deadlines; the twin replays measured curves for exact evaluation. Controllers compared on identical seeded traces, from uncapped through static and Joule Point caps and cap-only (the study's fixed-placement result) to the full joint controller, under fixed-budget, price, carbon and step-curtailment envelopes, isolating each decision's marginal value. Each epoch: a mixed-integer program with a convex core (the measured P(θ) law is convex over the operating range), solved by rolling-horizon decomposition, a greedy-with-regret assignment layer over an exact per-card convex capping subproblem, validated against the exact offline solution in the twin. When an envelope makes the full load infeasible, the controller defers or rejects lower-priority work and reports unmet demand, so flexibility carries an explicit, measured service cost. Structural results: the epoch problem is NP-hard; the convex core is solved exactly via an equal-marginal-energy characterization; the greedy layer is analyzed for an approximation guarantee on deadline-slack workloads.
WP4	**Grid flexibility and the Israel demonstration.** Formal flexibility products (envelope commitment, offer, delivery record) with compliance, ramp and recovery metrics, defined and judged at wall power; demonstrated on a rented BMC-metered GPU server, its out-of-band meter doubling as the compliance record, plus the twin at fleet scale under the Israeli firm-plus-flexible scenario. The facility-level optimum in the twin: measured node curves plus overhead (cooling, distribution losses) from published PUE ranges.
WP5	**Economics, policy and open release.** (a) Incentive model: a principal-agent model of renter, operator and grid, calibrated on the measured law and latency identity, the renter's action the power cap; derives the renter's optimum per billing structure and proves alignment conditions, formalizing that per-hour billing rewards the energy-worst point [1]. (b) Conditional-connection valuation: the flexible tranche priced as a curtailment option from the WP4-measured curtailment cost, published Noga curtailment statistics and the Authority's tariff book (TAOZ), anchored to the Authority's June 2026 hearing on a dedicated connection track for consumers of 50 MVA and above [19], an instrument the regulator is designing now. (c) Twin validation: every candidate instrument becomes an envelope rule evaluated in the WP3 twin on the same seeded traces, on measured curves; firm-plus-flexible scenarios of 500 to 2,000 MW quantified against the 27,000 MW queue. Public release of ELF-2, controller, twin and interfaces.

Success metrics are set in advance: law fit and Joule Point location per card and engine; transfer-model error on held-out cards; energy per served job and deadline compliance versus baselines; envelope violation, ramp and recovery in the demonstration; sensitivity of every economic figure to its price inputs.

2.5.5.6 Novelty and originality

- **AI load as a grid resource.** Sub-second actuation plus deadline-aware control turns an AI facility into a provider of measurable firm and flexible capacity, a category we have not found in practice or literature, immediately valuable in Israel's connection crisis.
- **From per-job search to one-time measurement, with an exact price.** The near-constant per-card cap turns what earlier systems compute per job at runtime into a one-time per-card-model measurement, at under one percent penalty [1]; and the extra cards needed to keep throughput equal the latency ratio, so every recommendation carries its exact capital cost.
- **Energy first, power as the means.** We measured the cost of universal full-power operation (roughly a third of inference energy) and located the alternative; existing capping satisfies watts while possibly raising joules, while JouleFlex sets power to the computed energy optimum per (card, job) and prices every departure.
- **From the board to the wall.** Published GPU energy results are board-power results; measuring the node's Joule Point at server-input power moves the claim toward what the grid actually meters.
- **The joint allocation-and-capping problem.** In the cited work, assignment and power control are solved separately; coupled through the shared power budget they define a new scheduling-control problem, and the controller is a first measured solution.

2.5.5.7 Fit to the Ministry's R&D goals

Operators own the knobs, the Ministry owns the rules, as a ministry sets fuel-economy standards without driving cars: JouleFlex gives the Ministry the measured basis for connection terms, planning coefficients and efficiency standards. The proposal addresses the Ministry's 2026 R&D policy [17] under call 34/2026 [18], in particular high-priority clauses of Chapter 2:

Clause	How the project addresses it
16.2 · טכנולוגיות תכנון וניהול הספקת אנרגיה לחוות שרתים	The core deliverable: power shaping manages a server farm's energy supply in real time; the flexibility framework tells planners how much a farm needs.
6.2 · טכנולוגיות לניהול אנרגיה מקומי וניהול ביקושים	A facility following a power envelope on instruction, at ~0.2-second actuation, is a demand-response resource of grid-relevant size.
8 · התייעלות באנרגיה וביזור מערכות אנרגיה (priority 2026)	29–31 percent less energy per AI task at the card, 18–45 percent less per served job at the fleet, in software on installed hardware.

16.1 · שימוש בבינה מלאכותית ככלי עזר לפיתוח ומחקר בכל הנושאים לעיל (2026 priority)	AI compute is the subject, learned models and control the instruments, an energy-system tool the deliverable.
15.3 · פיתוח מתודולוגיה לבניית תחזיות ומודלים ופיתוח כלי עזר בקבלת החלטות	Flexibility products, compliance metrics and the regulator-facing report are decision-support instruments for connection policy.

2.5.5.8 Preliminary results

The preliminary results are the applicant's study *The Joule Point* [1] (under review) and its ELF dataset: about 5,500 instrumented measurements over 20 models and four GPUs establish the response law (median $R^2 = 0.99$), locate the Joule Point at 43 to 46 percent of TDP with a 29 to 31 percent energy saving per task at 1.2 times longer runs, show one static cap per card serving all twenty workloads at under one percent mean penalty, and demonstrate an 18 to 45 percent fleet-level saving at equal-or-better deadline compliance in a measured-curve simulation, with about 50 percent more work per megawatt behind a fixed GPU power budget. Cap actuation settles in about 191 milliseconds. Under the per-instance-hour billing model analyzed in [1], the renter's cost optimum is the energy-worst operating point. ELF spans four GPU types; the fine sweeps behind the law and the Joule Point cover three (A100, A10G, L4), with the T4 swept coarsely for coverage, and $R^2 = 0.99$ is the median per fixed-setting sweep while pooled per-card fits across workloads are 0.88 to 0.92 (Fig. A1). The benefit persists under tight service: in the strict-deadline experiment of [1], capping saved 28 percent of energy while keeping 95 percent of workload value. The harness, simulator and analysis pipeline ran end to end; the project scales them out rather than building from zero.

2.5.6 Application: expected benefit and impact

2.5.6.1 What the project delivers, concretely

The project delivers five artifacts: the ELF-2 open dataset, the characterization kit, the JouleFlex controller, the digital twin, and the grid interface with the policy report; Appendix Table A1 states what each is and who uses it. Each artifact is released openly on delivery, every release crediting the Ministry's R&D program under call 34/2026 [18].

2.5.6.2 Benefit and impact

For operators: the preliminary results indicate 29–31 percent lower energy per task (measured) and 18–45 percent lower per served job (fleet simulation), through software control on installed hardware.

For the Israeli grid and regulator: the larger benefit. The binding constraint is connection capacity: about 27,000 MW of pending requests [2] strain planned generation through 2035.

The Joule Point converts efficiency into capacity, about 50 percent more AI work per connected megawatt; power shaping lets facilities accept *conditional* connections: firm plus flexible capacity curtailed on instruction. With flexibility measured, committed and audited (WP4), the regulator gains a third option and the frozen queue stops being all-or-nothing. The measured capacity gain is a GPU-board-level result; as an illustrative upper bound, every 1,000 MW of the queue connected under Joule Point operation could deliver the AI work of up to 1,500 MW uncapped, and a workload that would consume 1 TWh uncapped would avoid about 0.3 TWh, some 150,000 tonnes of CO_2 at 0.5 kg per kilowatt-hour; WP1 and WP4 measure how much of the gain survives at server input and facility level before any connection coefficient is recommended. The regulator-facing report quantifies this option and will be submitted as a formal response to the Authority's open consultation on the flexible-connection track [19], placing measured numbers in the docket while the rules are written; the open release lets any operator or authority adopt or certify the mechanism.

Training of manpower. The project trains a doctoral researcher at the machine-learning-systems and power-engineering intersection, a combination Israel produces in small numbers; beyond the funded researcher, ELF-2 and the twin become teaching infrastructure in the PI's HIT courses, with final-year projects built on the toolkit, training cohorts rather than one student. Results are published in peer-reviewed venues.

2.5.7 Work plan

2.5.7.1 Stages, deliverables, duration and success metrics

Months	Stage	Deliverable	Success metric
M1–M6	WP1: harness; new-generation and production-engine (vLLM) sweeps; three-tier instrumentation	D1: ELF-2 v1	Law fit $R^2 \geq 0.95$ per sweep or deviation documented; Joule Point located (or absence documented) per platform, board and wall
M5–M8	WP1: shared GPUs, LLM phases, fine-tuning; sweeps where the constant should break	M1: regime map	Documented validity domain of the law and the constant; where a job model is required
M7–M12	WP2: job model; transfer model; one-test characterization protocol	M2: models result	Job-model error versus the constant, per regime; transfer-model placement within 5 percentage points of TDP on held-out cards; protocol cost ≤ 8 hours and ≤ ₪1,500 per card model
M9–M16	WP3: twin; joint controller under fixed,	M3: controller result	Energy per served job and deadline compliance versus uncapped, static and cap-only baselines; marginal value of joint assignment; zero

	price, carbon and curtailment envelopes		exceedance beyond the WP1-calibrated tolerance in committed windows, overshoot magnitude and duration reported; transient violations within tolerance in fewer than 1 percent of epochs
M14–M20	WP4: flexibility products and metrics; demonstration on rented BMC-metered servers; Israel scenario	M4: flexibility demonstration	Sheddable power as a fraction of the committed envelope, measured on a BMC-metered server, projected to fleet scale in the twin; ramp within 60 seconds, recovery within 5 minutes; delivery-record compliance and curtailment service cost
M16–M21	WP5: incentive model; connection valuation on published Israeli data; twin evaluation	M5: policy report draft	Alignment condition per billing structure, confirmed in the twin; option value per MW and break-even curtailment frequency, with sensitivity to every price input
M21–M24	WP5: release, documentation, validation, publication	D2: open release; D3: final report	Public repositories with reproducing pipelines; metered validation; final scientific and policy reports

Two interim briefings to the chief scientist unit are named deliverables: an M8 technical brief and the M21 policy report draft before publication, in Hebrew, circulable to the Authority and Noga while the flexible-connection track [19] is under design. Every milestone is an inspectable artifact tested against pre-stated criteria; a bounding result is itself a deliverable. Minimum-success path: H100/L40S-class validation, one production serving engine, server-input BMC measurement, the job and transfer models, the joint controller, and one Israeli flexibility scenario; B200-class cards, a second engine, fine-tuning, shared-GPU regimes and the formal approximation guarantee are extension objectives pursued as access and time allow.

2.5.7.2 Budget by stage

Stage	Share	Amount (₪)	Principal costs
M1–M8 (WP1; WP2 begins)	30%	149,010	Doctoral researcher; GPU rentals for cap sweeps; instrumentation
M9–M16 (WP2 ends; WP3; WP4 begins)	33%	163,911	Doctoral researcher; card-model rentals for transfer validation; twin and controller work
M17–M21 (WP4 ends; WP5)	25%	124,175	Doctoral researcher; BMC-metered rentals; flexibility campaign
M22–M24 (WP5: release and validation)	12%	59,604	Metered validation runs; documentation; dissemination and publication
Total		**496,700**	Full itemization in the מפרט תקציבי

2.5.7.3 Risks and mitigation

Risk	Impact	Mitigation
Production serving software (continuous batching, shifting phases) blurs the per-card constant	Static cap loses near-optimality for LLMs	Measured first (WP1, M1); the job model predicts the optimum where the constant breaks
Power-cap control unavailable on some rented platforms	Measurement campaign narrowed	AWS EC2 gives passthrough and root; too-high cap floors are exposed via the graphics clock (L4 [1]); BMC rentals cover the wall tier
No grid partner for the Israel scenario	Demonstration less realistic	The scenario runs on public Noga and Authority data; a letter is additive, not blocking
The split incentive blocks adoption	Impact confined to first-party operators	WP5 targets the incentive structure; first-party operators and regulated connections adopt first

2.5.8 The applicant

Project manager and principal investigator: Alexander Apartsin, Senior Lecturer, Computer Science, HIT; single-PI, responsible for all five work packages.

Fit of expertise to the proposed work. The scientific foundation is the PI's own study [1]: the ELF dataset, the law, the Joule Point, the cost identity, the fleet saving. The work plan maps onto that record: power sweeps (WP1); response models with quantified uncertainty (WP2); simulators and schedulers against measured ground truth (WP3, WP4); techno-economic analysis with its incentive analysis already in [1] (WP5). The harness, models and simulation the work scales are built and validated; open release enables independent scrutiny. **Research personnel.** One doctoral researcher (90% position, both years) carries the measurement, modeling and control work under the PI's supervision, reported within a month of signature (נספח טז' §2.5).

Resources available to the applicant. The project owns no hardware: board sweeps run on AWS EC2 with root access, the RAPL tier on AWS bare metal, and the wall tier and WP4 demonstration on hourly dedicated servers with BMC PSU telemetry. HIT provides laboratory space, computing and grant administration; the sole purchase is one reference meter. **Collaboration:** an Israeli operator or grid body is sought for WP4 (§5.7 letter if secured).

2.5.9 Funding

No other funding exists or is applied for; per נספח טז' §1.5.2 and §2.10, no complementary funding will be received from any other body and no student receives parallel state funding.

2.5.10 Bibliography

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Appendix A: Figures (within the 5-page allowance of §2.2 נספח א׳1)

JouleFlex · קול קורא 34/2026 · Appendix to the research proposal, permitted under §2.2 נספח א׳1 (graphs, tables and diagrams, up to 5 A4 pages, outside the 10-page limit). All figures are measured results from the applicant's study [1].

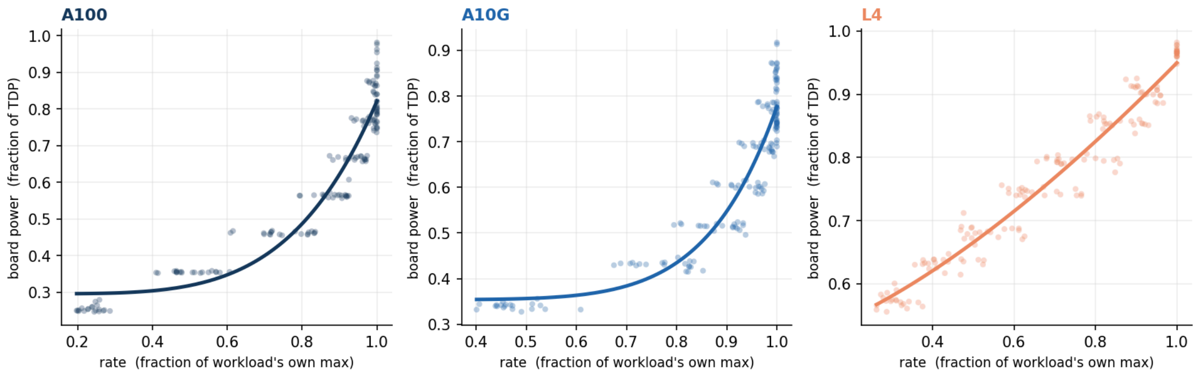

Figure A1. The response law is a card property. One panel per GPU (A100, A10G, L4); each dot is a measured operating point for one of the 20 workloads in the loaded regime, rate normalized to the workload's own maximum and power to the card's TDP. All workloads follow a single fitted law per card, $P(\theta) = P_0 + a\,\theta^\beta$ (pooled per-card R^2 of 0.92, 0.88 and higher per-sweep), showing the exponent travels with the card rather than the model. This is the empirical basis for the one-time per-card characterization that WP2 turns into a transfer model.

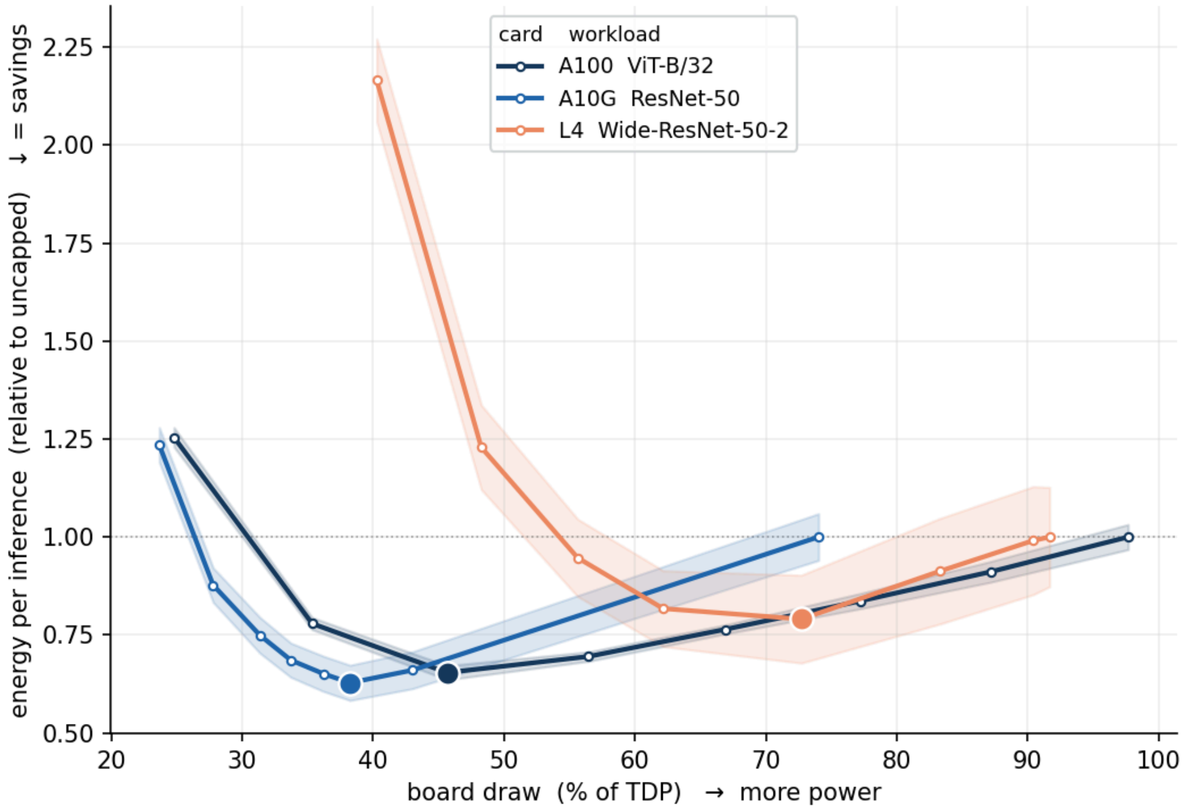

Figure A2. Every card has an energy U-shape. Energy per inference, normalized to each card's own uncapped value, against board draw as a fraction of TDP; the dot marks the Joule Point, the shaded band one standard deviation over repetitions. On the large GPUs the minimum sits at 43–46 percent of TDP and capping to it cuts energy per inference by 29–31 percent; on the L4 the interior minimum lies below the power cap's reachable floor and is exposed by the graphics clock instead, mapping which actuator applies to which card class.

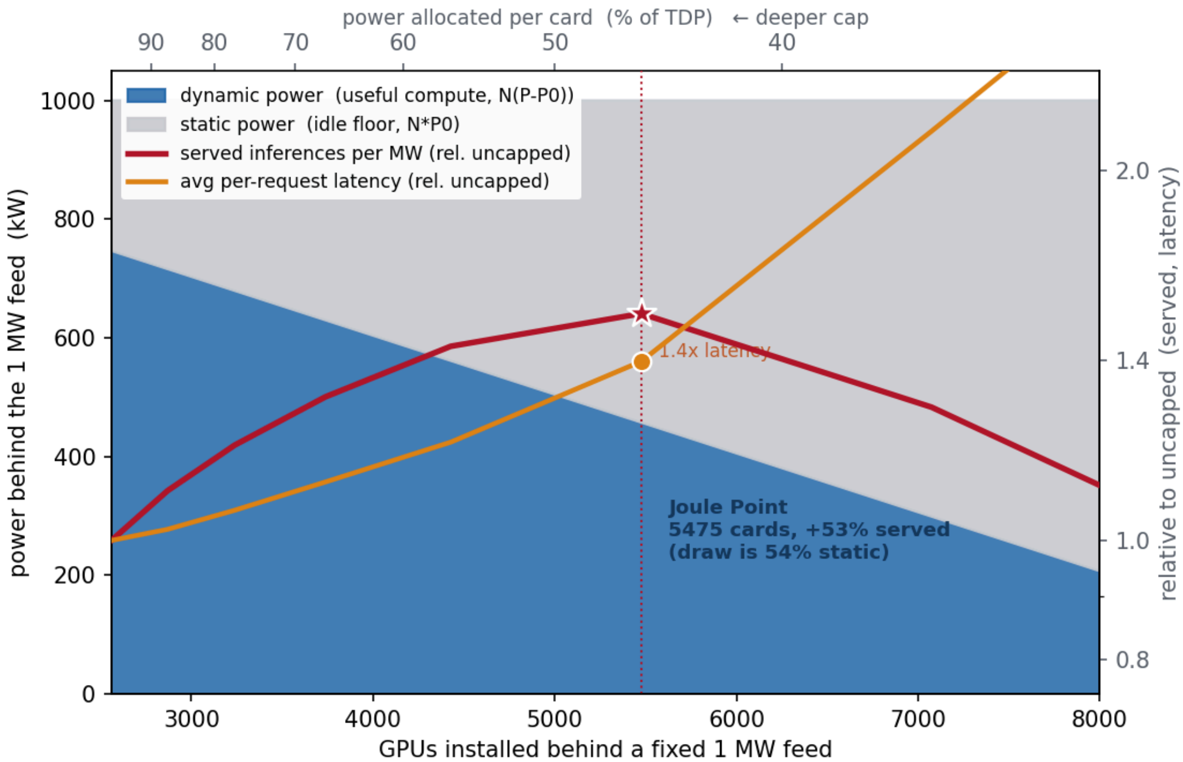

Figure A3. When power, not capital, binds, the Joule Point is the interior optimum. Behind a fixed 1 MW feed (A100, ViT-B/32 at batch 64), installing more GPUs forces each to a deeper cap, shifting the megawatt from useful dynamic compute (blue) to the static idle floor (grey). Served inferences per megawatt (red) peak at the Joule Point, near 5,500 installed cards, about 50 percent above the uncapped configuration. This is the capacity lever the Israel demonstration (WP4) operationalizes: under a rationed connection, efficiency becomes throughput. The same measured curves, allocated exactly across a fleet, dominate uncapped operation at every power budget from 100 down to 40 percent of total TDP; the closed-loop controller result appears in section 2.5.5.8.

Table A1. The five deliverables: what each is, who uses it

The project delivers five named artifacts, each with a defined user and use:

Deliverable	What it is	Who uses it, for what
ELF-2	An open dataset: power, throughput and latency at every operating point, board to wall, on current GPUs and production serving software	Researchers and vendors, as ground truth; every other deliverable is built and validated on it
The characterization kit	Software plus protocol: a one-test procedure and the two fitted models placing the Joule Point of any card, including unmeasured ones, in hours	A GPU engineer or operator qualifying a new card model or a new serving stack before deployment, at known cost
The JouleFlex controller	An open-source controller engine that assigns jobs and sets caps to keep a fleet inside its envelope, evaluated in the twin and demonstrated on a metered rented server before any deployment	An AI data-center manager, to cut energy per job and honor a committed envelope; its report doubles as the compliance record
The digital twin	A simulator replaying the measured curves of a described fleet and job mix, evaluating any policy or envelope scenario exactly before hardware is touched	An operator testing policies before deployment; a planner or regulator asking what a proposed facility could deliver and shed under a given connection
The grid interface and policy report	A blueprint: open formats for the envelope commitment, the flexibility offer and the delivery record, plus a regulator-facing report quantifying firm-plus-flexible connections against the national queue	The Electricity Authority and Noga, as the technical basis for offering conditional connections and auditing compliance

Table A1. The chain of use is direct: ELF-2 calibrates the models, the models feed the controller, the twin proves the controller before deployment, and the grid interface turns the controlled facility into something the regulator can contract with.